

Comprehensive Guide to MAD2 Viva Questions

Based on the previous viva questions, I've compiled detailed answers for the most commonly asked questions. This guide will help you prepare for your upcoming viva examination.

Vue.js Fundamentals

What is Vue.js?

Vue.js is an open-source JavaScript framework for building user interfaces and single-page applications. Created by Evan You, it focuses on the view layer of applications. Vue extends standard web technologies (HTML, CSS, and JavaScript) with an intuitive API that enhances their functionality. It's designed to be incrementally adoptable, allowing developers to integrate it into projects gradually rather than all at once.

What is a reactive framework?

A reactive framework is a programming model that deals with asynchronous data streams and enables applications to respond automatically to changes. It consists of two main components:

- **Reactive programming:** An asynchronous programming model that handles data streams, allowing easy expression of static or dynamic data.
- **Reactive systems:** An architectural style enabling microservices to communicate effectively. These systems are event-driven, interactive, flexible, resilient, and responsive through asynchronous message-driven communication.

Difference between v-if and v-show

- **v-if:** A conditional directive that completely adds or removes elements from the DOM based on a boolean expression. If the expression evaluates to false, the element won't be rendered at all. This is more efficient for elements that rarely change.
- **v-show:** Controls element visibility using CSS's display property (display:none). The element remains in the DOM regardless of the condition but is hidden when the expression is false. This is better for elements whose visibility changes frequently.

What are Vue lifecycle hooks and which ones have you used?

Vue lifecycle hooks are special methods that allow code execution at specific stages of a component's lifecycle:

- **beforeCreate:** Called before the instance is created

- **created:** Called after the instance is created but before mounting
- **beforeMount:** Called right before mounting the DOM
- **mounted:** Called after the DOM has been mounted
- **beforeUpdate:** Called when data changes, before the DOM is re-rendered
- **updated:** Called after a data change causes the DOM to be re-rendered
- **beforeUnmount:** Called before a component is unmounted
- **unmounted:** Called after a component has been unmounted

Common uses include data fetching in created, DOM manipulation in mounted, and cleanup in beforeUnmount.

What is v-cloak?

The v-cloak directive is used to hide uncompiled template content until Vue has finished compiling. It prevents users from seeing raw template syntax (like {{message}}) during page loading. It works by applying CSS rules that hide content with the v-cloak attribute until compilation is complete. This is particularly useful for Vue applications that compile in the browser rather than Single File Components.

Advanced Vue Concepts

What are props in Vue?

Props are custom attributes for passing data from parent components to child components. They establish a one-way data flow from parent to child, preventing child components from directly modifying parent data. Props can be defined with validation requirements including type checking, required flags, default values, and custom validators.

How does watch work in Vue and where would you use it?

The watch option in Vue allows you to observe and react to data changes. Unlike computed properties which are used for derived values, watchers are best for performing asynchronous or expensive operations in response to changing data. You would use watch when you need to:

- Make API calls when data changes
- Perform complex calculations that would be inefficient as computed properties
- React to specific data changes with side effects
- Validate input with debounced responses

What are slots in Vue?

Slots are a content distribution API in Vue that allows you to compose components in a way that parent components can pass content to child components. They enable flexible component composition and reusability. There are three types:

- Default slots: Basic content insertion
- Named slots: Multiple insertion points with names
- Scoped slots: Slots that can access child component data

What is Vuex?

Vuex is a state management pattern and library for Vue.js applications. It serves as a centralized store for all components in an application, with rules ensuring state can only be mutated in a predictable manner. It consists of:

- State: The central source of truth
- Getters: Computed properties for the store
- Mutations: Synchronous functions that modify state
- Actions: Asynchronous operations that commit mutations
- Modules: Way to divide the store into manageable chunks

What are mutations and actions in Vuex?

- **Mutations:** Synchronous functions that directly modify the state. They are the only way to change state in a Vuex store and must be committed using the commit method.
- **Actions:** Asynchronous operations that commit mutations after some process (like API calls) completes. Actions are dispatched using the dispatch method and can contain arbitrary asynchronous operations.

Backend Technologies

What is celery?

Celery is an asynchronous task queue/job queue based on distributed message passing. It's focused on real-time operations but also supports scheduling. Celery is used to execute tasks outside the HTTP request-response cycle, such as:

- Sending emails
- Generating reports
- Processing uploads
- Periodic tasks (like daily reminders)
- Resource-intensive computations

Components of celery

- **Workers:** Processes that execute tasks
- **Brokers:** Message queues (like Redis or RabbitMQ) that store tasks until a worker can process them

- **Result Backends:** Stores task results (optional)
- **Beat:** Scheduler for periodic tasks
- **Flower:** Web-based monitoring tool for Celery clusters

What is caching?

Caching is a technique that stores copies of frequently accessed data in a high-speed data storage layer to reduce data retrieval time. In web applications, caching:

- Improves application performance by retrieving data from fast memory instead of slower disk-based storage
- Reduces database load by serving repeated requests from cache
- Provides predictable performance during traffic spikes
- Eliminates database hotspots by caching frequently accessed data
- Increases read throughput (IOPS) significantly

Difference between memoization and caching

- **Memoization:** A specific optimization technique that stores results of expensive function calls and returns the cached result when the same inputs occur again. It's typically implemented at the function level and is a form of programmatic caching.
- **Caching:** A broader concept that involves storing data temporarily in a faster storage medium. Caching can occur at various levels (HTTP, database, application) and isn't necessarily tied to function inputs and outputs.

Difference between memcached and Redis cache

- **Memcached:**
 - Simple key-value store
 - No data persistence
 - No data structures beyond strings
 - Simpler to set up and use
 - Multi-threaded architecture
- **Redis:**
 - Advanced data structures (lists, sets, sorted sets, hashes)
 - Persistence options
 - Built-in replication
 - Transaction support
 - Lua scripting
 - Single-threaded but highly efficient

Security Concepts

What is a rainbow table?

A rainbow table is a precomputed table of reversed password hashes used to crack passwords in a database. It's a time-memory tradeoff technique that allows attackers to recover passwords from hashed values without having to compute the hash for every possible password. Rainbow tables are named for the different "colors" (hash and reduction functions) used in their creation. They're particularly effective against unsalted password hashes.

What is SHA-256? Can it be decrypted back?

SHA-256 (Secure Hash Algorithm 256-bit) is a cryptographic hash function that generates a fixed-size 256-bit (32-byte) hash value from input data of any size. Key characteristics:

- It's a one-way function, meaning it cannot be "decrypted" back to the original input
- Even a small change in input produces a completely different hash
- It's designed to be collision-resistant (difficult to find two inputs that produce the same hash)
- It's used for data integrity verification, digital signatures, and password storage

What is CSRF and CORS?

- **CSRF (Cross-Site Request Forgery):** An attack where unauthorized commands are transmitted from a user the website trusts. Protection methods include anti-CSRF tokens, SameSite cookies, and checking the Referer header.
- **CORS (Cross-Origin Resource Sharing):** A security feature implemented by browsers that restricts web pages from making requests to a different domain than the one that served the original page. It helps prevent unauthorized cross-origin access to resources.

Difference between authentication and authorization

- **Authentication:** The process of verifying who someone is. It confirms the identity of users through credentials like username/password, tokens, or biometrics.
- **Authorization:** The process of verifying what specific applications, files, and data a user has access to. It determines permissions and access rights to resources after authentication.

Database Concepts

What is ORM?

Object-Relational Mapping (ORM) is a programming technique that converts data between incompatible type systems in object-oriented programming languages and relational databases. Benefits include:

- Writing database operations in object-oriented code instead of SQL

- Abstracting database vendor-specific SQL
- Handling database connections and transactions
- Providing data validation and type conversion
- Implementing relationships between data

How to index in SQLAlchemy?

In SQLAlchemy, you can create indexes in several ways:

1. Using the Index construct:

```
from sqlalchemy import Index, Column, Integer, String, Table, MetaData

metadata = MetaData()
users = Table('users', metadata,
              Column('id', Integer, primary_key=True),
              Column('name', String),
              Column('email', String)
)
Index('idx_user_email', users.c.email)
```

2. Using the index parameter in Column:

```
class User(Base):
    __tablename__ = 'users'
    id = Column(Integer, primary_key=True)
    email = Column(String, index=True)
```

3. Using **table_args** for composite indexes:

```
class User(Base):
    __tablename__ = 'users'
    id = Column(Integer, primary_key=True)
    first_name = Column(String)
    last_name = Column(String)
    __table_args__ = (
        Index('idx_names', 'first_name', 'last_name'),
    )
```

What is ON DELETE CASCADE?

ON DELETE CASCADE is a referential action for foreign key constraints that automatically deletes rows in the child table when the corresponding rows in the parent table are deleted. This maintains referential integrity by ensuring that related records are consistently managed. In SQLAlchemy, it's implemented as:

```
class Parent(Base):
    __tablename__ = 'parent'
    id = Column(Integer, primary_key=True)
```

```
class Child(Base):
    __tablename__ = 'child'
    id = Column(Integer, primary_key=True)
    parent_id = Column(Integer, ForeignKey('parent.id', ondelete='CASCADE'))
```

Difference between primary key and unique key

- **Primary Key:**
 - Can only be one per table
 - Cannot contain NULL values
 - Automatically indexed
 - Identifies each record uniquely
 - Often used as a reference in foreign key relationships
- **Unique Key:**
 - Multiple unique constraints can exist in a table
 - Can contain NULL values (typically only one NULL)
 - Ensures uniqueness but doesn't serve as the main identifier
 - Can be used for alternate keys or business constraints

Web Technologies

What is AJAX?

AJAX (Asynchronous JavaScript and XML) is a set of web development techniques that allows web applications to send and retrieve data from a server asynchronously without interfering with the display and behavior of the existing page. Despite its name, AJAX can use JSON instead of XML, and it's not a programming language but a combination of:

- XMLHttpRequest object or Fetch API
- JavaScript and DOM manipulation
- HTML and CSS for display

AJAX enables more interactive web applications by:

- Updating parts of a web page without reloading the entire page
- Requesting data from a server after the page has loaded
- Receiving data from a server in the background
- Sending data to a server in the background

What is lazy loading?

Lazy loading is a design pattern that defers the loading of resources until they're actually needed. In web development, this typically refers to:

- Images that load only when they scroll into view
- JavaScript modules that are imported only when required
- Components that render only when they become visible

In Vue.js, lazy loading components can be implemented using dynamic imports:

```
const MyComponent = () => import('./MyComponent.vue')
```

This improves initial load time and performance by reducing the amount of code that needs to be downloaded, parsed, and executed upfront.

What is SPA (Single Page Application)?

A Single Page Application is a web application that loads a single HTML page and dynamically updates the content as the user interacts with it, without reloading the entire page. SPAs use AJAX and HTML5 to create fluid user experiences similar to desktop applications.

Characteristics include:

- Fast transitions between views
- Reduced server load (after initial load)
- Client-side rendering
- State management in the browser
- Routing handled by JavaScript

Vue.js, React, and Angular are popular frameworks for building SPAs.

Difference between PUT and PATCH HTTP methods

- **PUT:** A request method that updates an entire resource. If the resource doesn't exist, it may create it. PUT is idempotent, meaning multiple identical requests have the same effect as a single request.
- **PATCH:** A request method that applies partial modifications to a resource. It only updates the fields provided in the request, leaving others unchanged. PATCH is not necessarily idempotent.

Additional Concepts

What is a dataclass?

In Python, a dataclass is a class that is automatically equipped with special methods like **init()**, **repr()**, and **eq()** based on the class attributes. Introduced in Python 3.7, dataclasses reduce boilerplate code for classes that primarily store data. Features include:

- Automatic generation of **init()** and other special methods
- Default values for fields
- Type annotations
- Post-initialization processing
- Immutable instances option

How can you access the server without using browser and URL?

Several methods exist to access a server without a browser:

1. **Command-line tools:** curl, wget, httpie
2. **API clients:** Postman, Insomnia
3. **WebSockets:** For bidirectional communication
4. **SSH/Terminal access:** Direct server access
5. **Custom applications:** Using libraries like requests in Python
6. **Remote desktop protocols:** RDP, VNC, or web-based solutions like Apache Guacamole
7. **Mobile apps:** Native applications that communicate with servers

What is a webhook?

A webhook is a mechanism that allows one application to provide real-time information to another application as events occur. Unlike traditional APIs where you must poll for data, webhooks deliver data to other applications as it happens, making them more efficient for event-driven scenarios. Common uses include:

- Notification systems
- Payment processing updates
- Source code repository events (like GitHub commits)
- CRM updates
- IoT device state changes

Potential Additional Questions

What is the Virtual DOM in Vue.js?

The Virtual DOM is a lightweight JavaScript representation of the actual DOM. Vue uses it to minimize direct manipulation of the DOM, which is computationally expensive. When state changes occur, Vue:

1. Creates a new virtual DOM representation
2. Compares it with the previous one (diffing)
3. Calculates the minimal number of DOM operations needed
4. Applies only those changes to the real DOM

This approach significantly improves performance for complex UIs with frequent updates.

How would you handle form validation in Vue.js?

Form validation in Vue can be implemented through:

1. **Built-in validation attributes:** Using HTML5 attributes like required, minlength, pattern
2. **Custom validation with computed properties:** Computing validation status based on form data
3. **Watchers:** Validating data as it changes
4. **Form validation libraries:** Vuelidate or VeeValidate for more complex scenarios
5. **Custom validation components:** Creating reusable validation components

What security measures would you implement in a Vue.js + Flask application?

- **Frontend:**
 - Input validation and sanitization
 - XSS prevention (Vue automatically escapes content)
 - CSRF protection
 - Secure handling of tokens
 - Content Security Policy
- **Backend:**
 - Authentication and authorization
 - Input validation
 - Parameterized queries to prevent SQL injection
 - Rate limiting
 - HTTPS enforcement
 - Secure cookie settings
 - Proper error handling that doesn't expose sensitive information

How would you optimize the performance of your application?

- **Frontend:**

- Code splitting and lazy loading
- Proper use of v-show vs v-if
- Using keys with v-for
- Avoiding expensive operations in computed properties
- Implementing memoization for complex calculations
- Asset optimization (minification, compression)

- **Backend:**

- Implementing caching strategies
- Database query optimization
- Using appropriate indexes
- Asynchronous processing with Celery
- Horizontal scaling for high-traffic scenarios
- CDN integration for static assets

These comprehensive answers should help you prepare for your viva examination. Remember to relate these concepts to your specific project implementation when answering questions.

